ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
JAIPUR CENTER
Academic Session : 2021-22

SRGBT

for
JEE (MAIN+ADVANCED) 2022
ENTHUSIAST COURSE

Date : 25-07-2021 | Time : 3.00 Hr | CLASS-XII

Student's Form No. : ..... **Batch :** .....

Student's Name : .....

INSTRUCTIONS

1. **PART-1 : Physics** : **(Q. No. 1 to 4)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 5 to 10) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 11 to 18) : **Numerical Type Questions : Marking Scheme (+3, 0)**
2. **PART-2 : Chemistry** : **(Q. No. 19 to 22)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 23 to 28) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 29 to 36) : **Numerical Type Questions : Marking Scheme (+3, 0)**
3. **PART-3 : Mathematics** : **(Q. No. 37 to 40)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 41 to 46) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 47 to 54) : **Numerical Type Questions : Marking Scheme (+3, 0)**

PART-1 : PHYSICS

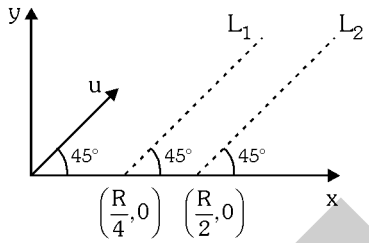
SINGLE CORRECT TYPE (+3, –1)

1. In a football game, a player wants to throw a ball to his one of teammate, who is running on the field. Taken thrower position as origin and receiver initial position at $2\hat{i}+3\hat{j}$, where $\hat{i}$ & $\hat{j}$ are in the plane of field. In subsequent run receiver displacement is $5\hat{i}$ then $8\hat{j}$, then $-2\hat{i}+4\hat{j}$ then $-6\hat{j}$. How far is receiver from thrower? (all displacements are in meter)

(A) $\sqrt{106}$ (B) $\sqrt{110}$ (C) $\sqrt{103}$ (D) $\sqrt{112}$

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2. A particle is projected with speed u from origin at an angle 45° with positive x-axis. There are two parallel imaginary lines L_1 and L_2 in the x-y plane as shown in figure. Line L_1 passes through $\left(\frac{R}{4}, 0\right)$ and line L_2 passes through $\left(\frac{R}{2}, 0\right)$. Lines are inclined at 45° with x-axis. Acceleration due to gravity is 'g' along negative y-axis. [where $R = \frac{u^2}{g}$].



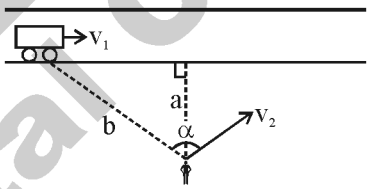
The angle θ between acceleration and velocity of particle between two instant of crossing the lines L_1 and L_2 is always :

(A) $\theta > 45°$ (B) $\theta < 45°$ (C) $\theta < 90°$ (D) $\theta > 90°$

3. The current velocity of a river grows in proportion to the distance from its bank and reaches the maximum value v_0 in the middle. Near the banks the velocity is zero. A boat is moving along the river in such a manner that it is always perpendicular to the current and the speed of the boat in still water is u. Find the distance through which the boat crossing the river will be carried away by the current if the width of the river is c.

(A) $\frac{cv_0}{u}$ (B) $\frac{cv_0}{2u}$ (C) $\frac{cv_0}{4u}$ (D) $\frac{cv_0}{8u}$

4. A bus is moving on a highway with a speed of $v_1 = 20$ m/s. A man is at a perpendicular distance of a = 60 m from the highway and b = 400 m from the bus. In order to catch the bus suppose he runs at an angle α from the line of joining him to the bus in initial condition. The man can run at a speed of $v_2 = 6$ m/s. Then the possible values of α so that he can reach any point on the highway at the same time as the bus or before it. Highway at the same time as the bus or before it.



(A) 160° (B) 180° (C) 15° (D) 120°

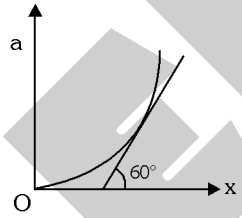
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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

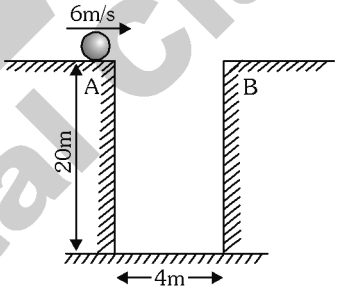
Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

5. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature as shown in figure. At $x = \sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at $x = \sqrt{3}$



(A) $v = \sqrt{(\sqrt{3}+9)}\ m/s$ (B) $a = 1.5\ m/s^2$ (C) $v = \sqrt{12}\ m/s$ (D) $a = 3\ m/s^2$

6. A small ball is moving with the speed of 6 m/s on horizontal surface falls in a vertical trench 4m-wide and 20m deep. Consider all collisions elastic and friction is absent every where. choose the correct statement(s).



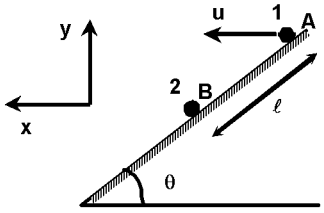
(A) The ball will come out of the trench at A (B) The ball will come out of the trench at B
(C) The ball takes time 4s time and comes out (D) The ball never comes out of it

7. A cylindrical pipe of radius r is at rest. A frog sitting on the horizontal ground jumps off and passes over the pipe touching it only at the top. Denoting air-time of the frog by T, horiozntal range of the jump by R and acceleration due to gravity by g, which of the following conclusions can you make ?

(A) $T = 4\sqrt{\frac{r}{g}}$ (B) $T = 2\sqrt{\frac{r}{g}}$ (C) $R = 2(\sqrt{gr})\sqrt{\frac{r}{g}}$ (D) $R = 4(\sqrt{gr})\sqrt{\frac{r}{g}}$

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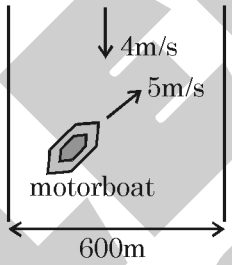
8. A smooth inclined plane has an angle of inclination θ from the horizontal. From point A on the inclined plane, body 1 is projected horizontally with speed u and from point B on the inclined plane, body 2 is released from rest. Distance AB is equal to ℓ. If the magnitude of relative velocity of body 2 w.r.t. body 1 and magnitude of relative acceleration of body 2 w.r.t. body 1 at time t are v and 'a', respectively, then which of the following is/are true?



(A) $v = \sqrt{u^2 - 2ug\sin\theta\cos\theta t + g^2\cos^2\theta t^2}$ (B) a = zero

(C) $a = g\cos\theta$ (D) $v = \sqrt{(g\sin^2\theta t - u)^2 + (g\sin\theta\cos\theta t - gt)^2}$

9. A 600 m wide river flows towards South at 4 m/s. A small motor boat travels at 5 m/s in still water & points in such a direction so that it will travel towards East relative to land.



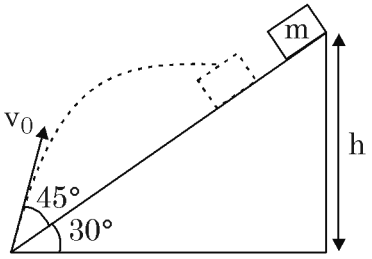
(A) The speed of boat relative to land is 3 m/s.

(B) The speed of boat relative to land is $\sqrt{41}$ m/s.

(C) The time taken by boat to cross the river is nearly 200 second.

(D) The time taken by boat to cross the river is nearly 100 second.

10. A particle of mass m is projected up from the bottom of an smooth inclined plane with initial velocity v_0 at angle 45° with an inclined plane of inclination 30° as shown in figure. At the same time a small block of same mass m is released from rest at a height h, the particle hits the block at some point on inclined plane. (take g = 10 m/s^2) (size of the block is negligible)



(A) The value of height h is $\frac{v_0^2}{\sqrt{3}g}$.

(B) The value of height h is $\frac{\sqrt{3}v_0^2}{g}$.

(C) If the particle stick to the block after collision, the velocity of block parallel to the inclined plane just after collision will be $\frac{v_0}{\sqrt{2}}$.

(D) If the particle stick to the block after collision, the velocity of block parallel to the inclined plane just after collision will be $\frac{v_0}{2\sqrt{2}}\left(\frac{4}{\sqrt{3}} - 1\right)$.

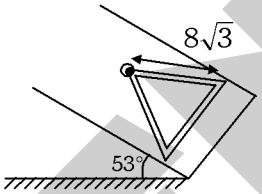
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NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

11. A child throws a ball so as to clear a wall of height h and at a distance x from it. Find the minimum speed required for clearing the wall is $\sqrt{g(ah+\sqrt{bh^2+cx^2})}$ then find a + b + c ?

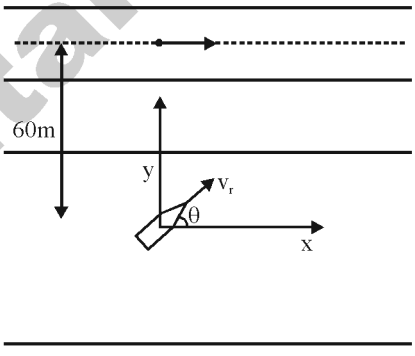
12. Two particles are projected from a point at the same instant with velocities whose horizontal and vertical components are u_1, v_1 and u_2, v_2 respectively. Prove that the interval between their passing through the other common point of their path is $\frac{a(bv_1u_2+cv_2u_1)}{g(u_1+u_2)d}$. Find a + b + c + d ?

13. A particle is released from the top of an equilateral triangular groove which is meet in the plane of an incline plane of inclination $\theta = 53°$. The corners of triangular groove are curved to neglect impulse. Side of triangle is $8\sqrt{3}$ m. Calculate the time of a complete round trip (at corners there is no change in speed).



14. Two tourists who are at a distance of 40 km from their camp must reach it together in shortest possible time. They have one bicycle which they decide to use in turn. One of them started walking at a speed of 5 km/hr and the other rod of on the bicycle at a speed of 15 km/hr. The tourists agreed to leave bicycle at intermediate points between which one walks and other ride. How long bicycle remain unused (in minutes) :

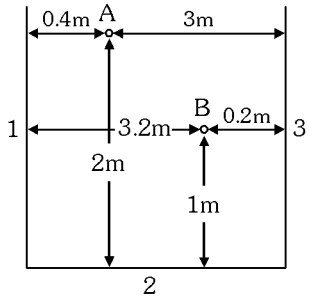
15. Figure shows top view of a boat moving with velocity $\vec{v}_r = \left(8\hat{i}+6\hat{j}\right)$ m/s relative to water. Water flows with velocity $\vec{v}_w = 2\hat{i}$ m/s relative to ground. A motor cyclist moves along a road running parallel to bank with velocity $\vec{v}_m = 20\hat{i}$ m/s. An apple thrown from boat is caught by motor cyclist in same horizontal plane after 4 sec of projection. If apple is thrown with velocity $v_x\hat{i}+v_y\hat{j}+v_z\hat{k}$ relative to boat fill value of $\frac{|\vec{v}_x|+|\vec{v}_y|+|\vec{v}_z|}{13}$.



16. A boy crosses a river twice on a straight path at an angle $\phi = 60°$ with the downstream direction, first time in two minutes andsecond time in four minutes. If his speed relative to river current is $v_{b/w} = \sqrt{3}$ m/s in both the attempts, find speed of the river current.

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17. Two point A and B are located on floor of a room as shown in figure. The room has three walls 1, 2 and 3. Perpendicular distance of point A from wall 1, 2, and 3 is 0.4 m, 2m and 3m respectively. While that of point B is 3.2 m, 1m and 0.2 m respectively. A man has to move from point A to B touching the walls 1, 2 and 3 at least once. Minimum distance (in meter) traveled by man will be :



18. The x-coordinate of a particle moving on x-axis is given by $x = 3 \sin 100t + 8 \cos^2 50t$, where x is in cm and t is time in seconds. Find the maximum distance the particle can move from the origin (in cm).

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PART-2 : CHEMISTRY

SINGLE CORRECT TYPE (+3, –1)

19. Calculate the sum of stoichiometirc coefficients of products in minimum integral balanced following chemical reaction.

$$a\ CrCl_3 + b\ H_2O_2 + c\ NaOH \longrightarrow x\ Na_2CrO_4 + y\ NaCl + z\ H_2O$$

(A) 8 (B) 16 (C) 45 (D) 32

20. 100 m*l* of a gaseous mixture containing Ne, CO_2 & H_2 on complete combustion in just sufficient amount of O_2 showed contraction of 60 ml at STP. When the resulting gases were passed through KOH solution, volume reduces by 40%. The volume ratio of $V_{CO_2} : V_{Ne} : V_{H_2}$ in original mixture is.

(A) 10 : 9 : 2 (B) 6 : 9 : 10 (C) 7 : 3 : 10 (D) None of these

21. The molality of pure gas A is $\frac{50}{2.4}$ and that of pure B is $\frac{100}{3.2}$. Calculate molarity of A in a gaseous solution of A and B at 8 atm pressure and $\frac{200}{0.821}$ K. if the vapour density of the gaseous mixture at 10 atm and 400 K is 18.

(A) 0.4 (B) 0.1 (C) 0.3 (D) 0.2

22. 100 ml of a sample of hard water requires 25.1 ml of 0.02 N H_2SO_4 for complete reaction. The hardness of water (density = 1 gm/ml) is :

(A) 200 ppm (B) 251 ppm (C) 261 ppm (D) 288 ppm

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

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23. 40 ml, 0.05 M solution of sodium sesquicarbonate dihydrate [$Na_2CO_3.NaHCO_3.2H_2O$] is titrated against 0.05 M HCl solution, x ml of acid is required to reach the phenolphthalein end point, while y ml. of same acid was required when methyl orange indicator was used in a separate titration. Which of the following statements are correct ?

(A) y – x = 80 ml

(B) y + x = 160 ml

(C) If the titration is started with phenolphthalein indicator and methyl orange is added at the end point, 2x ml. HCl would be required further to reach end point.

(D) y + x = 120 ml.

24. Which of the following options represent concentration terms which are independant of temperature ?

(A) % w/w, mole fraction, molality (B) Volume strength of H_2O_2, normality, ppm

(C) Molality, mass fraction, % labelling (D) Molarity, % w/v, ,mole fraction

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25. A definite volume of pure ammonia (NH_3) gas is passed through a series of electric sparks by which the volume becomes 90 ml. The increase in volume is due to formation of nitrogen (N_2) and Hydrogen (H_2)gases. All the gases finally present are washed with dilute sulphuric acid solution, by which the volume of gases become 80 ml. All the volumes are measured at the same temperature and pressure. Which of the following statement(s) is/are **correct** regarding the original ammonia sample?
(A) The volume of NH_3 gas taken was 40 ml.
(B) The volume of NH_3 gas taken was 50 ml.
(C) Only 80% of NH_3 gas decomposed into N_2 and H_2 gases.
(D) Only 20% of NH_3 gas decomposed into N_2 and H_2 gases.

26. Which of the following option/s is/are correct w.r.t to equivalent mass (EM).
(A) Equivalent mass of HCl in its reaction with $KMnO_4$ to give $MnCl_2$ & Cl_2 is equal to 36.5

(B) Equivalent mass of MnO_2 when it converts to Mn_2O_7 & Mn^{2+} in a molar ratio of 2 : 1 is $\frac{5}{6}$ times of its molecular mass

(C) Equivalent mass of NaOH in its reaction with S_8 to give Na_2SO_4 & Na_2S in $\frac{4}{3}$ times its molecular mass.

(D) Equivalent mass of $K_4[Fe(CN)_6]$ when it reacts with $FeCl_3$ to form $KFe[Fe(CN)_6]$ is $\frac{1}{3}$ of its molecular mass.

27. For the reaction

$$CrO_5 + H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + H_2O + O_2$$

Incorrect statement is/are :
(A) Oxidation state of 'Cr' in CrO_5 is +8. (B) Oxidation of one of element in CrO_5 takes place.
(C) Reduction of one of element in CrO_5 takes place. (D) Oxidation state of 'Cr' in CrO_5 is +10.

28. 1 mole of a mixture of CO & CO_2 requires exactly 1 litre solution of 1 M NaOH for complete neutralisation. If CO present in mixture is now converted to CO_2 and again the mixture is treated with NaOH, then after this conversion :
(A) moles of CO_2 present initially in mixture = 1.
(B) 2 litre NaOH solution of 1 M is more required for neutralisation.
(C) 2 litre solution of $\frac{M}{2}$ is more required for neutralisation.
(D) 56 gm KOH in aqueous solution is required more for neutralisation.

NUMERICAL TYPE

(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

29. Given balanced chemical equation for oxidation of phosphorus (III) sulfide by nitric acid. The products include NO and SO_2. Find the value of a, b, c, d, e in the following equation.

$$a\ P_4S_6 + 44H^+ + b\ NO_3^- \rightarrow cNO + dH_3PO_4 + eSO_2 + 4H_2O$$

find the value of (a + b + c + d + e)

30. 50 ml of gaseous mixture of acetylene and ethylene if taken in a ratio of a:b requires 700 ml of air containing 20% by volume O_2 for complete combustion. Calculate the volume of air required for complete combustion of a mixture having ratio b : a.

31.

Composition of Oleum			**Solution Code**
50 gm SO_3	;	120 gm H_2SO_4	5
230 gm SO_3	;	100 gm H_2SO_4	6
80 gm SO_3	;	98 gm H_2SO_4	2
133 gm SO_3	;	86 gm H_2SO_4	3
93 gm SO_3	;	93 gm H_2SO_4	1
400 gm SO_3	;	500 gm H_2SO_4	4

Mention the solution code of the Oleum having maximum % labelling.

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32. Consider the production of tetraethyl lead according to the following reaction.

$$4C_2H_5Cl + 4Na - Pb \longrightarrow (C_2H_5)_4Pb + 4NaCl + 3Pb$$

How much kilograms of ethyl chloride is required to produce enough tetraethyl lead (density $\frac{12.96}{1.29} g/ml$) needed for 500 litre of aviation fuel using 2 ml of tetraethyl lead per litre of fuel ? (Pb = 208)

33. 200 mL of 1 M HCl solution is mixed with 800 mL of 2M HCl solution. Calculate the molarity of final solution.

34. A mixture of formic acid and oxalic acid is heated in the presence of H_2SO_4.

$$HCOOH \longrightarrow H_2O + CO$$

$$H_2C_2O_4 \longrightarrow H_2O + CO_2 + CO$$

The produced gaseous mixture is collected and cooled. On its treatment with KOH solution, the volume of gaseous mixture decreases by one–tenth. Calculate the molar ratio of the two acid in original mixturel.[If the simplest ratio is a : b then find a + b].

35. In the following reactions if initially 100 moles of A were taken then calculate sum of moles of D and C after the reaction.

$$A \xrightarrow{40\%} 3B + D$$

$$2B \xrightarrow{50\%} 4C$$

36. If 246 g of a triatomic ideal gas having only one type of atoms is found to occupy 44.8 L at 2 atm and 546 K then what will be the atomic weight of the element ?

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PART-3 : MATHEMATICS

SINGLE CORRECT TYPE (+3, –1)

37. If $\sin^2 x - 2(a+1)\sin x + (a^2 + 2a - 4) > 0$ for all $x \in R$, then the complete set of values of a is $(-\infty, m) \cup (n, \infty)$.

Find the value of $[n] - [m]$

([.] denotes greatest integer function)

(A) 5 (B) 6 (C) 7 (D) 8

38. Let f(x) be a polynomial function of degree at most 5 which leaves remainder – 1 and 1 upon divison by $(x-1)^3$ and $(x+1)^3$ respectively . The value of f(2) is equal to

(A) $\frac{46}{15}$ (B) $\frac{-23}{8}$ (C) $\frac{-23}{4}$ (D) none of these

39. Sum of all the integral solutions of the equation

$\sqrt{\left(\log_3 \sqrt[3]{3x} + \log_x \sqrt[3]{3x}\right)\cdot \log_3 x^3} + \sqrt{\left(\log_3 \sqrt[3]{x/3} + \log_x \sqrt[3]{3/x}\right)\cdot \log_3 x^3} = 2$ is equal to

(A) 4 (B) 5 (C) 6 (D) 7

40. Let f(x) be a polynomial function of 9 degree and having leading coefficient 2009.

If $f\left(\frac{1}{r}\right) = \frac{1}{r}$ for r = 1, 2, 3, ..., 9 and $f\left(\frac{1}{10}\right) = \frac{m}{n}$, then $\frac{n - 10m}{10}$ is equal to

(m , n are coprime)

(A) 0 (B) 2010 (C) 20090 (D) none of these

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

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Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

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41. Let A denotes the sum of the roots of the equation $\frac{1}{5 - 4\log_4 x} + \frac{4}{1 + \log_4 x} = 3$.

B denotes the value of the product of m and n, if $2^m = 3$ and $3^n = 4$.

C denotes the sum of the integral roots of the equation $\log_{3x}\left(\frac{3}{x}\right) + (\log_3 x)^2 = 1$.

Which of the following alternative(s) is/are correct ?

(A) The value of A + B equals 8 (B) The value of B + C equals 6

(C) The value of A + C ÷ B equals 8 (D) The value of A + C ÷ B equals 5

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42. If the complete set of x satisfying $\log_{\log_2\left(\frac{x}{2}\right)}\left(x^2-10x+22\right)>0$ is $(a,b)\cup(c,\infty)$, then which of the following alternative(s) is/are incorrect ?

(A) $a+b=8-\sqrt{3}$ (B) $a+c=8+\sqrt{3}$ (C) b + c = 10 (D) a + c = 10

43. Let $f(x)=-5x^3+ax^2+bx+c$ where $f(1)=1,f(2)=8,f(3)=27,f(4)=\lambda$. Also $\mu=\frac{\log_2 24}{\log_{96}2}-\frac{\log_2 192}{\log_{12}2}$ then which of the following alternative(s) is/are incorrect ?

(A) $\lambda-\mu=25$ (B) $\lambda+\mu=37$ (C) $\lambda>30$ (D) $\mu>3$

44. If $\left(\frac{\sqrt{a+4}}{1-a}-1\right)x^4-5<0$ for all $x\in R$, then the complete set of values of a is $[m,n]\cup(p,\infty)$.

Which of the following alternative(s) is/are correct ?

(A) $4m+(3-2n)^2-3p=2$ (B) $2m+(3-2n)^2-p=13$

(C) $(3-2n)^2+m=17$ (D) $(3-2n)^2+p=21$

45. If the inequality, $2\log_{0.5}y^2-3+2x\log_{0.5}y^2-x^2>0$ is valid for atleast one real value of 'x' , then y can have the value

(A) $\cot\left(\frac{29\pi}{15}\right)$ (B) $\tan\left(\frac{23\pi}{8}\right)$ (C) $\tan\left(\frac{46\pi}{15}\right)$ (D) $\cot\left(\frac{9\pi}{8}\right)$

46. If $2^x=49^{x-1}$, then x is equal to :

(A) $\frac{2\log_2 7}{\log_2 7-1}$ (B) $\frac{1}{1-\log_7\sqrt{2}}$ (C) $\frac{2}{2+\log_7\frac{1}{2}}$ (D) $\frac{1}{1+\log_{49}2}$

NUMERICAL TYPE

(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

47. Let $a=\left(\sqrt{\log_2\sqrt[4]{10}+\log_5\sqrt[4]{10}}-\sqrt{\log_2\sqrt[4]{\frac{5}{2}}+\log_5\sqrt[4]{\frac{2}{5}}}\right)\left(\sqrt{\log_2 5}\right)$,

$b=\left(\sqrt{\log_3\sqrt[4]{6}+\log_2\sqrt[4]{6}}-\sqrt{\log_3\sqrt[4]{\frac{2}{3}}+\log_2\sqrt[4]{\frac{3}{2}}}\right)\left(\sqrt{\log_3 2}\right)$.

Find antilog of (a + b) to the base 3

48. Let s be the sum of all the integers which are less than 10 and are satisfying the equation $|x^2-5x+6|-9=|x^2-6x|+|3-x|$, and p be the sum of all the integers which are less than 7 and are satisfying the equation $|x^2-5x+6|=|x^2-x|-|6-4x|$. Find the value of (s + p)

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49. Let $a = \left(\frac{\log_3\left(\log_{2^2} 5\right)}{\left(2\log_9\left(\left(3^{\log_{15} 30}\right)\left(5^{\log_{15} 2}\right)\right)\right)-1} \right)$, then $4^{\left(\sqrt{2}\right)^a}$ is equal to

50. Find the absolute value of sum of all the integral values of k for which the equation $|x+1|-|x|+3|x-1|-2|x-2|-2 = x+k$ has more than two solutions.

51. If any solution of the inequality, $\frac{\log_3\left(x^2-3x+7\right)}{\log_3(3x+2)} < 1$ is also a solution of the inequality $x^2 + (5-2a)x \le 10a$, then the least value of a is λ. Find the value of $(4\lambda - 3)$

52. Let P denotes number of digits in $8^{12}\ 5^{35}$ and Q denotes number of zeroes after decimal before a significant figure start in $\left(\frac{8}{27}\right)^{20}$ where $\log_{10}2 = 0.3010$ & $\log_{10}3 = 0.4771$. The value of (P + Q) equals.

53. Let $a = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - (\log_2 12 \cdot \log_2 48) + \log_2 64$, then find the value of $\sum_{k=0}^{\infty} \frac{a^k}{a^{a^k}+1}$

54. Let $a = \frac{\left(\log_2 3 - \log_5 7\right)\left(\log_2 3 - \log_7 9\right)}{\left(\log_3 5 - \log_5 7\right)\left(\log_3 5 - \log_7 9\right)}$, $b = \frac{\left(\log_2 3 - \log_3 5\right)\left(\log_2 3 - \log_7 9\right)}{\left(\log_5 7 - \log_3 5\right)\left(\log_5 7 - \log_7 9\right)}$

and $c = \frac{\left(\log_2 3 - \log_3 5\right)\left(\log_2 3 - \log_5 7\right)}{\left(\log_7 9 - \log_3 5\right)\left(\log_7 9 - \log_5 7\right)}$. Find the value of [a + b + c]

([.] denotes greatest integer function)

* * * * *

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SIDE 2

JEE (MAIN & ADVANCED)
(ANSWER SHEET)

FORM NO.

CANDIDATE NAME

PART - 1

SECTION - I

1	A	B	C	D	11	A	B	C	D
2	A	B	C	D	12	A	B	C	D
3	A	B	C	D	13	A	B	C	D
4	A	B	C	D	14	A	B	C	D
5	A	B	C	D	15	A	B	C	D
6	A	B	C	D	16	A	B	C	D
7	A	B	C	D	17	A	B	C	D
8	A	B	C	D	18	A	B	C	D
9	A	B	C	D	19	A	B	C	D
10	A	B	C	D	20	A	B	C	D

SECTION - II

Q1 (+ -) Q2 (+ -)

Q3 (+ -) Q4 (+ -)

Q5 (+ -) Q6 (+ -)

Q7 (+ -) Q8 (+ -)

SECTION - III

Q.NO.	1	2	3	4	5	6	7	8	9	10
	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4	4	4
	5	5	5	5	5	5	5	5	5	5
	6	6	6	6	6	6	6	6	6	6
	7	7	7	7	7	7	7	7	7	7
	8	8	8	8	8	8	8	8	8	8
	9	9	9	9	9	9	9	9	9	9

SECTION - IV

1(A)	P	Q	R	S	T	2(A)	P	Q	R	S	T
(B)	P	Q	R	S	T	(B)	P	Q	R	S	T
(C)	P	Q	R	S	T	(C)	P	Q	R	S	T
(D)	P	Q	R	S	T	(D)	P	Q	R	S	T

PART - 2

SECTION - I

1	A	B	C	D	11	A	B	C	D
2	A	B	C	D	12	A	B	C	D
3	A	B	C	D	13	A	B	C	D
4	A	B	C	D	14	A	B	C	D
5	A	B	C	D	15	A	B	C	D
6	A	B	C	D	16	A	B	C	D
7	A	B	C	D	17	A	B	C	D
8	A	B	C	D	18	A	B	C	D
9	A	B	C	D	19	A	B	C	D
10	A	B	C	D	20	A	B	C	D

SECTION - II

Q1 (+ -) Q2 (+ -)

Q3 (+ -) Q4 (+ -)

Q5 (+ -) Q6 (+ -)

Q7 (+ -) Q8 (+ -)

SECTION - III

Q.NO.	1	2	3	4	5	6	7	8	9	10
	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4	4	4
	5	5	5	5	5	5	5	5	5	5
	6	6	6	6	6	6	6	6	6	6
	7	7	7	7	7	7	7	7	7	7
	8	8	8	8	8	8	8	8	8	8
	9	9	9	9	9	9	9	9	9	9

SECTION - IV

1(A)	P	Q	R	S	T	2(A)	P	Q	R	S	T
(B)	P	Q	R	S	T	(B)	P	Q	R	S	T
(C)	P	Q	R	S	T	(C)	P	Q	R	S	T
(D)	P	Q	R	S	T	(D)	P	Q	R	S	T

PART - 3

SECTION - I

1	A	B	C	D	11	A	B	C	D
2	A	B	C	D	12	A	B	C	D
3	A	B	C	D	13	A	B	C	D
4	A	B	C	D	14	A	B	C	D
5	A	B	C	D	15	A	B	C	D
6	A	B	C	D	16	A	B	C	D
7	A	B	C	D	17	A	B	C	D
8	A	B	C	D	18	A	B	C	D
9	A	B	C	D	19	A	B	C	D
10	A	B	C	D	20	A	B	C	D

SECTION - II

Q1 (+ -) Q2 (+ -)

Q3 (+ -) Q4 (+ -)

Q5 (+ -) Q6 (+ -)

Q7 (+ -) Q8 (+ -)

SECTION - III

Q.NO.	1	2	3	4	5	6	7	8	9	10
	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4	4	4
	5	5	5	5	5	5	5	5	5	5
	6	6	6	6	6	6	6	6	6	6
	7	7	7	7	7	7	7	7	7	7
	8	8	8	8	8	8	8	8	8	8
	9	9	9	9	9	9	9	9	9	9

SECTION - IV

1(A)	P	Q	R	S	T	2(A)	P	Q	R	S	T
(B)	P	Q	R	S	T	(B)	P	Q	R	S	T
(C)	P	Q	R	S	T	(C)	P	Q	R	S	T
(D)	P	Q	R	S	T	(D)	P	Q	R	S	T

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

SRGBT-1

ENTHUSIAST COURSE

DATE : 25-07-2021

ANSWER KEY

PART-1 : PHYSICS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	B	D	AB	AC	AD	AC	AC	AD
Que.	11	12	13	14	15	16	17	18		
Ans.	3	3	5	160	3	1	5	9		

PART-2 : CHEMISTRY

Que.	19	20	21	22	23	24	25	26	27	28
Ans.	B	B	B	B	ABC	AC	BC	CD	AD	CD
Que.	29	30	31	32	33	34	35	36		
Ans.	121	675	6	8	1.8	9	160	41		

PART-3 : MATHEMATICS

Que.	37	38	39	40	41	42	43	44	45	46
Ans.	C	C	B	D	ABC	BC	BCD	AC	ABC	BC
Que.	47	48	49	50	51	52	53	54		
Ans.	6	50	25	2	7	46	1	1		